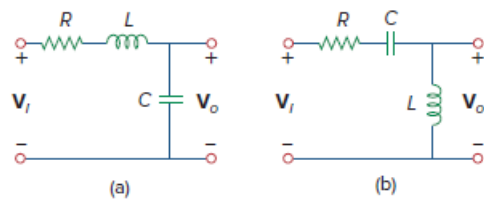


Problem

For each of the circuits in Fig. 10.70, find V_o/V_i for $\omega = 0$, $\omega \rightarrow \infty$, and $\omega^2 = 1/LC$.

Figure 10.70:



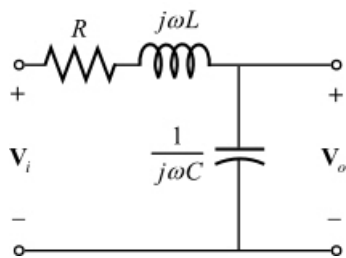
Step-by-step solution

Step 1 of 10

(a)

Refer to circuit diagram in Figure 10.70 (a) in the textbook.

Draw the frequency domain circuit.



Step 2 of 10

Calculate the transfer function.

$$V_o = \left(\frac{\frac{1}{j\omega C}}{R + j\omega L + \frac{1}{j\omega C}} \right) V_i$$

$$\frac{V_o}{V_i} = \frac{\frac{1}{j\omega C}}{\frac{j\omega RC - \omega^2 LC + 1}{j\omega C}}$$

$$= \frac{1}{j\omega RC - \omega^2 LC + 1}$$

[Comments \(1\)](#)



Anonymous

Though it may be assumed, it is helpful to note that voltage divider was applied

Step 3 of 10

Calculate V_o/V_i for $\omega = 0$.

$$\begin{aligned}\left.\frac{V_o}{V_i}\right|_{\omega=0} &= \frac{1}{j\omega RC - \omega^2 LC + 1} \\ &= \frac{1}{j(0)RC - (0)^2 LC + 1} \\ &= 1\end{aligned}$$

Therefore, the value of V_o/V_i for $\omega = 0$ is $\boxed{1}$.

Step 4 of 10

Calculate V_o/V_i for $\omega = \infty$.

$$\begin{aligned}\left.\frac{V_o}{V_i}\right|_{\omega=\infty} &= \frac{1}{j\omega RC - \omega^2 LC + 1} \\ &= \frac{1}{j(\infty)RC - (\infty)^2 LC + 1} \\ &= \frac{1}{\infty} \\ &= 0\end{aligned}$$

Therefore, the value of V_o/V_i for $\omega = \infty$ is $\boxed{0}$.

Step 5 of 10

Calculate V_o/V_i for $\omega^2 = \frac{1}{LC}$.

$$\begin{aligned}\left.\frac{V_o}{V_i}\right|_{\omega=\frac{1}{\sqrt{LC}}} &= \frac{1}{j\omega RC - \omega^2 LC + 1} \\ &= \frac{1}{j\left(\frac{1}{\sqrt{LC}}\right)RC - \left(\frac{1}{\sqrt{LC}}\right)^2 LC + 1} \\ &= \frac{1}{jR\sqrt{\frac{C}{L}} - 1 + 1} \\ &= \frac{1}{jR\sqrt{\frac{L}{C}}}\end{aligned}$$

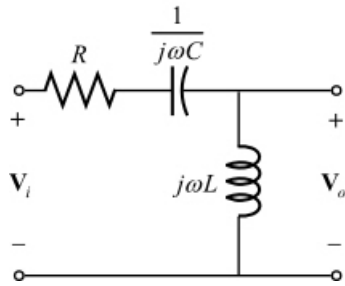
Therefore, the value of V_o/V_i for $\omega^2 = \frac{1}{LC}$ is $\boxed{\frac{1}{jR\sqrt{\frac{L}{C}}}}$.

Step 6 of 10

(b)

Refer to circuit diagram in Figure 10.70 (b) in the textbook.

Draw the frequency domain circuit.



Step 7 of 10

Calculate the transfer function.

$$\mathbf{V}_o = \left(\frac{j\omega L}{R + j\omega L + \frac{1}{j\omega C}} \right) \mathbf{V}_i$$

$$\begin{aligned} \frac{\mathbf{V}_o}{\mathbf{V}_i} &= \frac{j\omega L}{j\omega RC - \omega^2 LC + 1} \\ &= \frac{(j\omega L)(j\omega C)}{j\omega RC - \omega^2 LC + 1} \\ &= \frac{-\omega^2 LC}{j\omega RC - \omega^2 LC + 1} \end{aligned}$$

Step 8 of 10

Calculate $\mathbf{V}_o/\mathbf{V}_i$ for $\omega = 0$.

$$\begin{aligned} \left. \frac{\mathbf{V}_o}{\mathbf{V}_i} \right|_{\omega=0} &= \frac{-\omega^2 LC}{j\omega RC - \omega^2 LC + 1} \\ &= \frac{-(0)^2 LC}{j(0)RC - (0)^2 LC + 1} \\ &= 0 \end{aligned}$$

Therefore, the value of $\mathbf{V}_o/\mathbf{V}_i$ for $\omega = 0$ is 0.

Step 9 of 10

Calculate V_o/V_i for $\omega = \infty$.

$$\begin{aligned}\left.\frac{V_o}{V_i}\right|_{\omega=\infty} &= \frac{-\omega^2 LC}{j\omega RC - \omega^2 LC + 1} \\ &= \frac{-LC}{j\frac{RC}{\omega} - LC + \frac{1}{\omega^2}} \\ &= \frac{-LC}{j\frac{RC}{\infty} - LC + \frac{1}{\infty}} \\ &= \frac{-LC}{0 - LC + 0}\end{aligned}$$

$$= 1$$

Therefore, the value of V_o/V_i for $\omega = \infty$ is $\boxed{1}$.

Step 10 of 10

Calculate V_o/V_i for $\omega^2 = \frac{1}{LC}$.

$$\begin{aligned}\left.\frac{V_o}{V_i}\right|_{\omega=\frac{1}{\sqrt{LC}}} &= \frac{-\omega^2 LC}{j\omega RC - \omega^2 LC + 1} \\ &= \frac{-\left(\frac{1}{\sqrt{LC}}\right)^2 LC}{j\left(\frac{1}{\sqrt{LC}}\right)RC - \left(\frac{1}{\sqrt{LC}}\right)^2 LC + 1} \\ &= \frac{-1}{jR\sqrt{\frac{C}{L}} - 1 + 1} \\ &= \frac{j}{R}\sqrt{\frac{L}{C}}\end{aligned}$$

Therefore, the value of V_o/V_i for $\omega^2 = \frac{1}{LC}$ is $\boxed{\frac{j}{R}\sqrt{\frac{L}{C}}}$.